MIDI Drum Player (MDP) v1.0

Quick Start Guide

1. Introduction

The MIDI Drum Player (MDP) is designed to receive MIDI note data and play samples of percussion instruments. The MIDI data may come from a hardware sequencer or controller, a software sequencer, a MIDI file player, or any other source of MIDI percussion data.

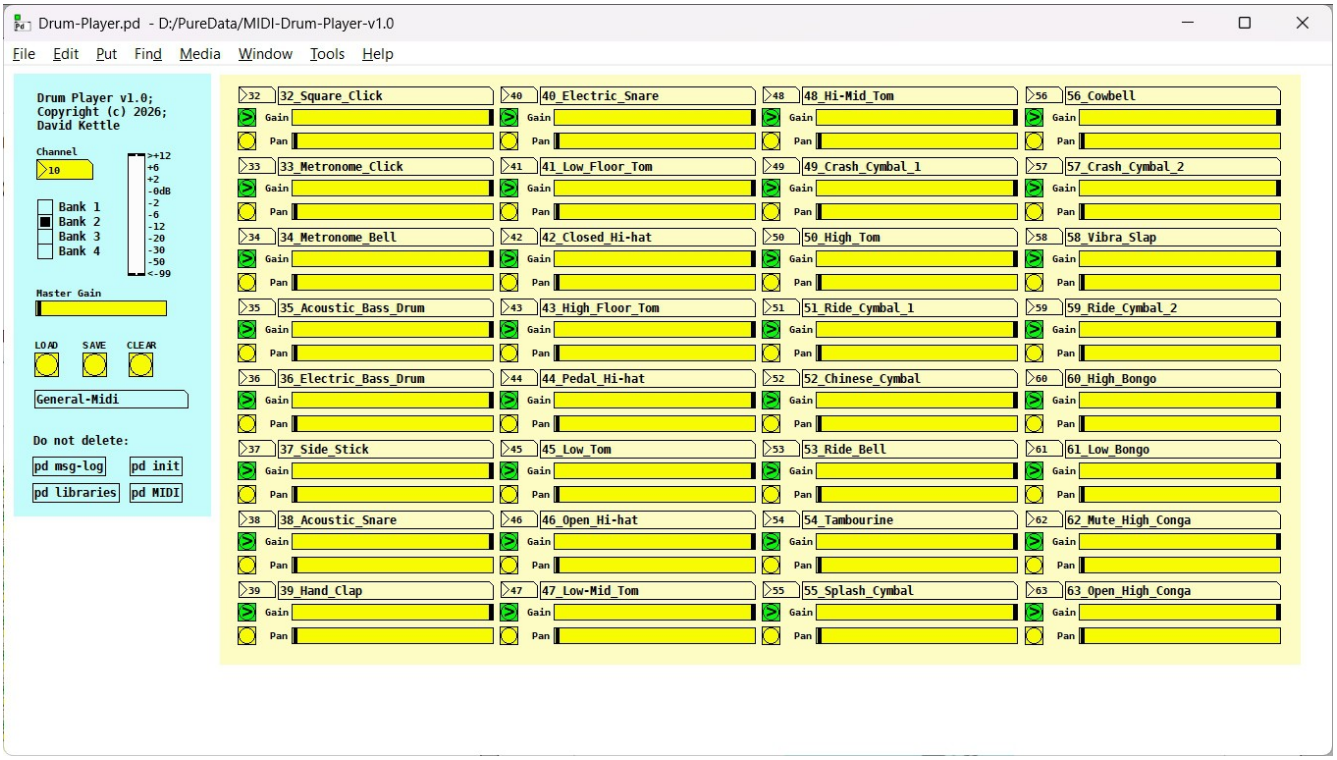


Fig. 1, Main Window

2. Installation Requirements

MDP v1.0 was developed and tested under Pure Data v0.56.2 (64-bit) and Windows 11. There's a good chance it will work on Linux and MacOS too, although it hasn't been tested.

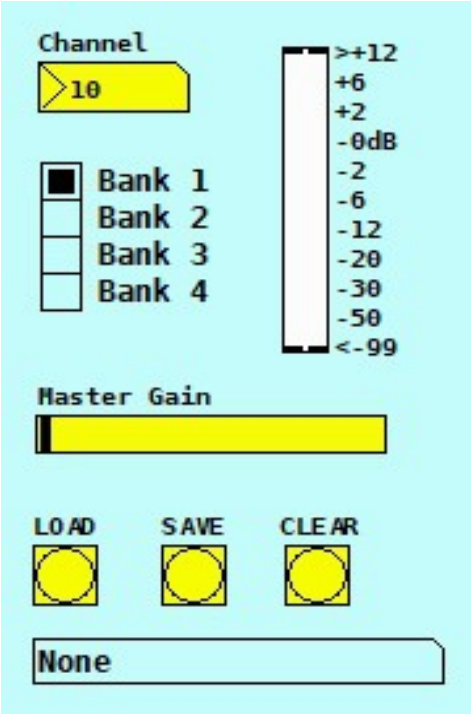
Although it may run under earlier versions of Pure Data, there's no guarantee, and you may encounter problems if you're running a much older version of Pure Data or any of the required external libraries.

The following libraries, all of which can be installed using Deken (under the Help menu, click on “Find externals”), are required:

- tof
- cyclone
- ggee
- zexy
- iemlib
- easyflow

3. Sound Banks

Fig. 2, Sound Banks



When you start up MDP (either by double-clicking on the file “Drum-Sequence.pd” or starting Pure Data and opening the file), the main window will open (see Fig. 1).

Sound banks can contain up to 128 monaural sample files in “wav” or “aiff” format, corresponding to MIDI note numbers 0 to 127 inclusive. Any unused notes should be set to “Unassigned”.

The panel on the left (see Fig. 2) is used to select the MIDI channel used for percussion (which defaults to 10), load/save/clear the sound banks, select a sound bank (from 1 to 4), and set the output volume (or gain).

To load a set of sound banks, click on “LOAD” and select a file containing the list of samples in the sound banks. Default sound banks are provided for General MIDI percussion instruments, but you can create your own sound banks that correspond to the hardware or software you are using.

To create a new set of sound banks, first click on “CLEAR” to clear all the instruments currently selected. Then select the sample files you want to include (see section 4, “Selecting Sample Files”) and click on “SAVE” to save the list under a different name.

The name of the currently-loaded set of sound banks is displayed underneath the LOAD/SAVE/CLEAR buttons.

Each set of sound banks is divided into 4 groups containing 32 instruments each. To display a group, click on one of the radio buttons labeled “Group 1” through “Group 4”.

4. Sample Files



Fig. 3, Loading and playing samples

To load individual sample files, click on the yellow button in the bottom left-hand corner of the instrument you wish to load (see Fig. 3). A window will open up allowing you to

navigate to the folder where you have saved your sample files and select one of them. After you click on “Open”, the sample file will be loaded and the name displayed at the top. If you decide not to load another sample file, click on “Cancel”.

You can listen to the sample you have loaded by clicking on the green button above the load button. Use the horizontal slider to the right of the green play button to set the volume or gain. If you don't hear anything, check that the main gain slider in the left panel is set sufficiently high.

The sample can by panned within the stereo field using the horizontal slider labelled “Pan” beneath the Gain slider.

5. Starting Playback

Once you have loaded the sound banks you want to use and selected the individual sample files, you can adjust the gain and panning of each instrument by using the sliders for that instrument. You can also adjust the overall volume by using the slider in the left panel. You can select which MIDI channel (from 1 to 16) your software or hardware device uses by setting the number in the “Channel” box in the left panel. Then start playback in the MIDI software or hardware device you are using, or start playing if you are performing live.

Note that the actual MIDI device to be used as input is specified under “Media / MIDI Settings...” in the Pure Data menu bar. Your MIDI device must be connected before starting Pure Data (if it's a hardware drum machine) or already running (if it's a soft synth running on your computer).

5. Getting Support

If you run into problems or have questions, please visit the project repository on Github:

https://github.com/dfkettle/MIDI-Drum-Player

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Appendix 1: General MIDI Percussion Map

27 High Q
28 Slap
29 Scratch Push
30 Scratch Pull
31 Sticks
32 Square Click
33 Metronome Click
34 Metronome Bell
35 Acoustic Bass Drum
36 Electric Bass Drum
37 Side Stick
38 Acoustic Snare
39 Hand Clap
40 Electric Snare
41 Low Floor Tom
42 Closed Hi-hat
43 High Floor Tom
44 Pedal Hi-hat
45 Low Tom
46 Open Hi-hat
47 Low-Mid Tom
48 Hi-Mid Tom
49 Crash Cymbal 1
50 High Tom
51 Ride Cymbal 1
52 Chinese Cymbal
53 Ride Bell
54 Tambourine
55 Splash Cymbal
56 Cowbell
57 Crash Cymbal 2
58 Vibra Slap
59 Ride Cymbal 2
60 High Bongo
61 Low Bongo
62 Mute High Conga
63 Open High Conga
64 Low Conga
65 High Timbale
66 Low Timbale
67 High Agogô
68 Low Agogô
69 Cabasa
70 Maracas
71 Short Whistle
72 Long Whistle
73 Short Guiro
74 Long Guiro
75 Claves
76 High Woodblock
77 Low Woodblock
78 Mute Cuica
79 Open Cuica
80 Mute Triangle

81 Open Triangle
82 Shaker
83 Jingle Bell
84 Belltree
85 Castanets
86 Mute Surdo
87 Open Surdo

Appendix 2: Sound Bank Files

Sound bank files can be added by creating text files in the following format. Each line should be terminated with a semi-colon. The first line specifies which MIDI channel to use. This is followed by 128 groups of 3 lines each, specifying gain, pan and the location of the sample file in WAV or AIFF format. Each of these lines contain 4 columns, specifying the parameter ('pan', 'gain' or 'path'), the bank no. (0 – 3), the slot no. (0 – 31) and the gain amount (0 – 1), pan amount (0 – 1) or the path (absolute or relative) to the sample file. If no sample file is to be loaded in a given slot, enter “Unassigned”.

For example:

```
channel 10;
pan 0 0 0.5;
gain 0 0 0;
path 0 0 Unassigned;
pan 0 1 0.5;
gain 0 1 0.776119;
path 0 1 ../data/TRADITIONAL_BRAZILIAN_PERCUSSION/AFOXE_HITS/Afoxe_Afoxe.wav;
pan 0 2 0.5;
gain 0 2 0.820895;
path 0 2 ../data/TRADITIONAL_BRAZILIAN_PERCUSSION/AFOXE_HITS/Agogo_1_A_Afoxe.wav;
pan 0 3 0.5;
gain 0 3 0.791045;
path 0 3 ../data/TRADITIONAL_BRAZILIAN_PERCUSSION/AFOXE_HITS/Agogo_2_A_Afoxe.wav;

(many lines omitted)

pan 3 31 0.5;
gain 3 31 0;
path 3 31 Unassigned;
```